P R O J E C T H A N D O F F

Quran Clause and Phrase Annotation

Merged Project Handoff — reference-backed annotation dataset with English and Bangla reader segmentation

> Supersedes: quranic_phrase_clause_handoff (A) and mulk-dataset HANDOFF (B)
>
> Pilot: Surah 67 Al-Mulk — Target: 114 surahs, 6,236 ayahs, 77,432 words
>
> Storage: canonical SQLite (references only) + rendered JSON views (cache)
>
> Next step: Phase 1 — source registry and read-only adapters

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1. Purpose of this document

This is the merged handoff for the Quran clause and phrase annotation project. It supersedes the two earlier handoffs: Package A (quranic_phrase_clause_handoff: reference-only annotation architecture over EQTB and QAC) and Package B (mulk-dataset: the Al-Mulk clause/phrase dataset with English and Bangla reader segmentation). The two packages were written at the end of separate working sessions; this document reconciles them into one project plan after the merge decisions were confirmed by the project owner.

The canonical, repo-ready version of this handoff is the Markdown file MERGED_HANDOFF.md; this Word document is a reading copy of the same content. A fresh session should read this document top to bottom and start work at the roadmap in Section 11. All figures quoted here were verified against the two packages: the Surah 67 production summary of Package A, and the built dataset of Package B (30 ayahs, 333 words, 213 phrases, 86 clauses).

2. Project goal

Build one Quran-wide clause and phrase annotation project that serves two coordinated purposes from a single canonical store. The reader layer (Package B's mission) segments every ayah into clauses and phrases, the minimal semantic building blocks that make natural sense, with English and Bangla for every phrase, such that joining the phrases of a clause in Arabic word order reads naturally in both languages; it must also support mining of repetition frequencies and segmentation rules afterwards. The linguist layer (Package A's mission) attaches, as an overlay, phrase and clause classifications across several dimensions with full provenance: every unit traceable to immutable source records, every classification carrying evidence and review status.

The pilot is Surah 67 (Al-Mulk). The target is the whole Quran: 6,236 ayahs, 77,432 QPC words, 114 surahs. The example below shows the core idea on 67:15.

Table 1 Example segmentation of 67:15 (Arabic order; en/bn per phrase)

Arabic phrase	English	Bangla
هُوَ ٱلَّذِي	He is the One who	তিনিই যিনি
جَعَلَ لَكُمُ ٱلۡأَرۡضَ ذَلُولٗا	made the earth manageable for you,	তোমাদের জন্য যমীনকে সুগম,
فَٱمۡشُواْ فِي مَنَاكِبِهَا	so walk through its regions	অতএব তোমরা বিচরণ কর
وَكُلُواْ مِن رِّزۡقِهِۦۖ	and eat from His provision;	তাঁর দেয়া রিযিক থেকে;

Arabic phrase	English	Bangla
وَإِلَيۡهِ ٱلنُّشُورُ	and to Him is the resurrection.	আর তাঁরই কাছে; পুনরুত্থান।

The five clauses tile the ayah: every word belongs to exactly one phrase, every phrase to exactly one clause, in Arabic word order.

3. Settled decisions (do not re-litigate)

The following decisions are settled. Items marked Merge were decided in the merge session by the project owner; items marked A or B are carried from the respective handoffs and remain binding there.

Table 2 Settled decisions

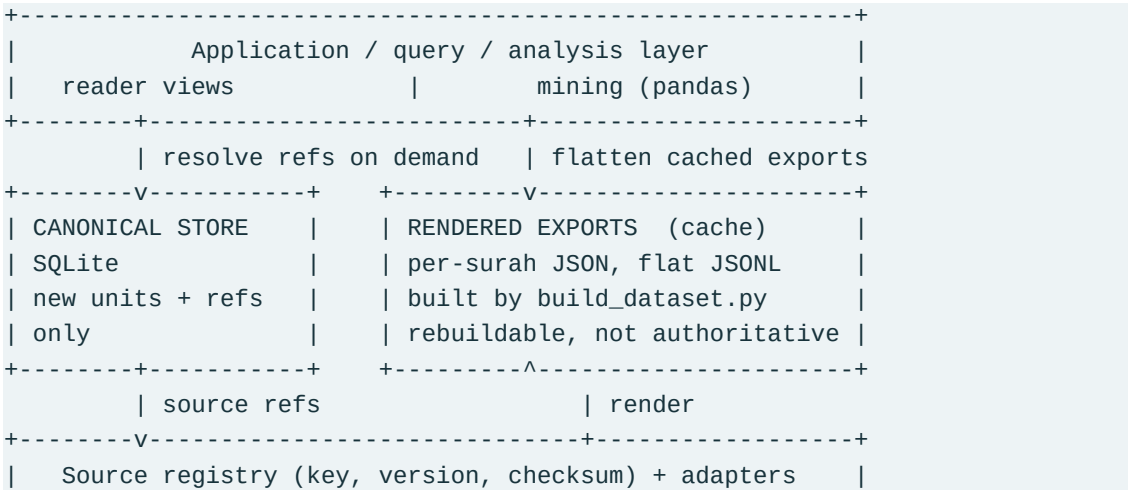
#	Decision	Origin
1	Hybrid storage: a canonical SQLite store holds only genuinely new annotations plus source references; per-surah JSON exports with inline morphology are rebuildable rendered views, explicitly marked as cache.	Merge
2	Package B's tiling segmentation is the base unit model; Package A's classification dimensions are an optional overlay on the same units.	Merge
3	Status vocabulary: candidate / pending / reviewed / accepted / uncertain / rejected (Appendix A maps B's three states onto them).	Merge
4	Both source families are canonical references: QUL for text, translations, glosses and morphology; EQTB (treebank) for constituents, sentence ids and relations; QAC morphology as cross-check.	Merge
5	First build step: source registry and read-only adapters.	Merge
6	JSON only for authoring and exports (CSV dropped); one nested JSON per surah; analysis flattens with pandas in memory.	B
7	Source priority: QUL first, Quranic Treebank second. GPL licence is acceptable.	B
8	Bangla reference translation is Zakaria (respectful আপনি forms). Rawai al-Bayan is dropped (uses তুমি).	B
9	Bangla phrases follow Arabic word order; no separate Bangla reading order.	B
10	Annotation must be independent of rules (anti-circularity): the annotating model sees only Arabic words, glosses, the two reference translations, CONVENTIONS.md and a glossary; tags and ids attach afterwards; rules are mined only after the dataset exists; translation memory flags conflicts only, never auto-fills.	B

#	Decision	Origin
11	Reference, never copy: existing datasets are immutable and independently versioned; no text, morphology, roots, lemmas, POS, dependencies, codebooks or lexicons are stored in the canonical store; source values resolve at render time.	A
12	Never force a single label: structural type, grammatical function, speech act and rhetorical function are separate, overlapping dimensions.	A
13	Uncertainty is explicit: automatically generated candidates are never exposed as gold annotations.	A + B
14	QPC word keys (s:a:w) are the canonical word identifiers; never tokenize QPC text by whitespace (37:130:3 contains a space inside one word).	B (promoted)
15	Dropped: QSU (kept finding: it split at the major pause mark 5/6 but the minor mark 0/9), recitation timings, dependency-relation storage (deferred), the mustafa0x morphology fork (replaced by converted treebank JSON).	B
16	Generation uses the user's OpenRouter API key through the existing pipeline.	B

Proposals that were pending in Package B are resolved by this merge as follows: word-range identifiers are adopted (Section 5.4); per-ayah status is adopted using the merged vocabulary; per-surah key-alignment validation becomes a Phase 1 exit criterion; CONVENTIONS.md stays as shared annotator memory; flat JSONL per level is deferred to Phase 2; the waqf_strength field idea (major vs minor pause) remains open.

4. Architecture: hybrid storage

The merged system has four parts: immutable reference sources, a canonical annotation store, a rendering pipeline, and the application and analysis layer on top. The canonical store is the released annotation dataset; everything else is derived or referenced.



```
|   read-only; resolvers: QPC key <-> EQTB <-> QAC         |
+----------------------------------------------------------+
| QUL files | EQTB Quranic.csv + aux | QAC | Mutashabihat |
+----------------------------------------------------------+
```

Five rules govern this architecture:

1. The canonical store is the released annotation dataset. It contains new phrase and clause units, their token-span references, phrase translations in English and Bangla (genuinely new content), classifications, relationships, evidence, reviews, and the glossary. It contains no word text, morphology, roots, lemmas, POS, dependency edges, ayah translations, word glosses or codebook rows; those resolve through adapters.
2. Rendered exports are cache. Package B's mulk.json shape stays as the reader format, but morphology, glosses and translations arrive resolved from sources at build time, and the file header records the source versions and checksums used. Any cache must be rebuildable from store plus sources alone.
3. Phrase translations live in the canonical store, because they are the project's own annotation. Full-ayah translations (Sahih International, Zakaria) and word glosses are source content: referenced by key, resolved at render.
4. If a phrase or clause already exists exactly as an EQTB constituent, reference that constituent (exact_entity role) instead of creating a duplicate unit. Create new units only for genuinely missing boundaries or relationships, storing minimum token-span references.
5. A source reference carries the dataset key, version and, where possible, a checksum, so that a changed release never silently repoints to different records.

5. Annotation model

5.1 Base layer: tiling segmentation

Definitions, kept from Package B's CONVENTIONS.md: a **phrase** is the smallest run of consecutive Arabic words that forms a natural, meaningful unit, not a word-by-word gloss; keep together what only makes sense together (a preposition with its noun, a possessive construction, a verb with its attached pronoun, a fixed expression). A **clause** is a consecutive run of phrases forming a clause-sized idea (statement, condition, result, relative clause, question, command, quoted speech act), decided by meaning rather than length. Tiling is strict: phrases cover words 1 to N of the ayah exactly once, in order, with no gaps or overlaps, and clauses tile the ayah the same way; phrases always follow Arabic word order.

Each base unit stores: its canonical identifier, its level (phrase or clause), its word span as QPC keys, the English and Bangla translations, a normalized **translation_key** (lower-case,

no punctuation, referent-neutral: “and it is” maps to the key “and he is” when the referent is Allah), source evidence references, and annotator, version and status. Consistency rules already applied on Al-Mulk: the same Arabic phrase always receives the same English and Bangla wording and the same translation_key unless the referent genuinely differs. Known normalizations: فَسَتَعْلَمُونَ becomes “So you will soon know”; فَمَن becomes “who then”; إِنَّهُۥ becomes “Indeed, He is”; and وَهُوَ in 67:4 stays “and it is” because it refers to the vision, while sharing the key “and he is”.

5.2 Overlay layer: linguistic classifications

Optional per-unit classifications span six dimensions: phrase_structure, clause_structure, grammatical_function, speech_act, rhetorical_function and discourse_function. Typed relationships (condition_of / result_of, subordinate_to, relative_clause_of, coordinates_with and similar) link units; a relationship is never encoded as a label on a single unit. For a conditional construction the condition and the result are separate clause units linked by a condition-result relationship; an EQTB CS span covers the larger construction and does not automatically supply both child spans.

The reuse-versus-create rule from Package A is binding:

1. Search existing sources for the same entity, span and annotation meaning.
2. If the exact information already exists and is adequate, reference it; do not copy or recreate it.
3. If the source has the entity but not the new classification, attach only the new classification to the source reference.
4. If the source lacks the boundary, entity or relationship, create a new annotation unit with source-token references only.
5. If the source meaning is ambiguous or a mapping is unvalidated, preserve the source-specific label and mark any crosswalk provisional.

Nesting and overlap policy: the reader layer is flat tiling by design. Nesting, overlap and multi-segment spans live in the overlay and in source constituents (EQTB constituents nest and overlap; never flatten them), resolved at runtime. An EQTB sentence may cross an ayah boundary; never split sentence-level structures at verse boundaries.

5.3 Status vocabulary

Canonical statuses are: candidate (auto-generated, unreviewed), pending (awaiting review), reviewed (checked by an annotator), accepted (approved into the released set), uncertain (plausible but unresolved) and rejected (reviewed and not accepted). A candidate must never be exposed as a verified gold label. Statistics and rule mining use reviewed and accepted items only. Appendix A maps Package B's generator statuses onto this vocabulary.

5.4 Identifiers

- Word: the QPC key s:a:w, 77,432 words in total. In qpc-hafs-word-by-word.json the last key of each ayah is the end-of-ayah marker, not a word.
- Unit identifiers (adopted from Package B's proposal): {surah}:{ayah}:{level}{start}-{end}, for example phrase 67:15:p3-6 and clause 67:15:c3-9. Cross-ayah overlay spans use {s}:{a}:{level}{start}-{s2}:{a2}:{end}. The legacy c2 / p1 counters remain as display labels.
- Source record keys: EQTB token tid and (sentence_id, token_id); EQTB constituent = sentence_id plus the inclusive constituents_loc token range; QAC location s:a:w:segment; QUL = file plus key. Every stored reference also carries a source_dataset_id combining key and version.

6. Sources (canonical references)

All sources are read-only references. The registry pins each one with dataset key, version or release, URL, licence identifier and file checksum.

Table 3 Source inventory

Source	File(s)	Role
QUL word list	quran-script/wbw/qpc-hafs-word-by-word.json	Master word list: keys s:a:w, QPC Hafs text, waqf marks. Authority on word counts.
QUL ayah text	digital-khatt-v2 script file	Ayah text for display.
QUL translations	en-sahih-international-simple.json; bn-dr-abu-bakr-muhammad-zakaria-simple.json	Reference ayah translations (English; Bangla with respectful forms).
QUL word glosses	english-wbw-translation.json; bangali-word-by-word-translation.json	Drafting hints for the annotating model only.
QUL morphology	grammar/word-root.db, word-lemma.db, word-stem.db	Root, lemma, stem (join *_words.word_location to the id table).
EQTB main	Quranic.csv, about 50 MB, UTF-16, tab-delimited (Mendeley rk96pn66m4/1; github.com/NoorBayan/Quranic)	Constituents, sentence ids, dependency relations; 139,376 rows; fields include tid, sentence_id, token_id, location, uthmani_token, pos, features, lemma, root, rel_label, ref_token_id, is_constituent, constituents_loc, constituent_label.

Source	File(s)	Role
EQTB auxiliary (in Package A)	RelLabels.csv (140), CAPoS.csv (45), ConstituentsTags.csv (6), CAMorphFeatures.xlsx (13 sheets), CALemmaLexicon.csv (4,833), CARootLexicon.csv (1,643); all UTF-16 tab	Codebooks and lexicons that interpret main-table values; not six more token corpora.
QAC morphology	quranic-corpus-morphology-0.4.txt	Independent morphology cross-check; locations s:a:w:segment.
Treebank JSON	quranic-treebank/data/surah-NNN.json, converted by convert_treebank.py, QPC-keyed	Tags, sentence ids and constituent spans for the build.
Mutashabihat	phrases.json, phrase_verses.json, matching-ayah.json	Repetition cross-check.
QUL metadata	metadata/quran-metadata-ayah.json	words_count is WRONG for 37:130 (counts 4; QPC and treebank give 3). Do not rely on it.

Licensing: the EQTB software licence does not establish the licence of its data files; QAC has specific data-use requirements; QUL terms are published at qul.tarteel.ai. Verify all three before any redistribution and keep licence metadata in the source registry; the annotation layer's own licence is chosen independently after that review. The full Quranic.csv and the QAC morphology file are deliberately not included in either package and must be supplied from the original downloads.

7. Data gotchas (merged, complete)

7.1 Scaling gotchas (Package B)

1. QPC writes the phrase بَعْدَ مَا as two words in 2:181, 8:6 and 13:37, but the treebank merges it into one word, so all later treebank words in those ayahs are numbered one lower. convert_treebank.py fixes this with the ALIGN table; output keys are QPC keys and the script asserts 77,432 words; the --qpc option checks every ayah. The English and Bangla gloss files keep QPC keys but lack the three second-part entries.
2. 37:130:3 إِلْ يَاسِينَ is ONE QPC word containing a space: never tokenize QPC text by whitespace; use word keys.
3. English WBW has 83,665 gloss keys, which equals the 83,668 QPC keys (77,432 words plus 6,236 end markers) minus the three missing second parts. The Bangla

gloss file lacks the end-marker key 2:181:15 (harmless) and contains stray quote marks and informal forms, so Bangla glosses are drafting hints only.

4. QUL versus QAC roots: 0 disagreements on 209 words of Surah 67, compared after removing spaces and treating the three alif variants alike. Lemmas differ on 28 words, mostly spelling (dagger alif and similar); three words have no QUL lemma (67:8:9, 67:19:9, 67:23:10); three words of the amman pattern are lemmatized min by QUL against am by QAC (67:20:1, 67:21:1, 67:22:7). Keep the qac_lemma field for comparison.
5. The treebank and QAC English descriptions in the VerbMood and VerbForm sheets of CAMorphFeatures.xlsx are wrong; do not use those sheets.

7.2 EQTB structure facts (Package A)

1. constituents_loc values such as [3-18] are inclusive, sentence-local token_id ranges; the lookup key is (sentence_id, token_id), not the Quranic location alone. The tid field also exists and can be retained as an additional record identifier.
2. An EQTB sentence may cross an ayah boundary; never split sentence-level structures at verse boundaries.
3. Implicit tokens (54 of them in Surah 67) have sentence_word = 0, location = underscore, and surface placeholders such as (هُوَ) or (*). They are legitimate source records with no Quranic word coordinates: never invent coordinates for them and exclude them from direct QAC matching.
4. EQTB constituent tags are PP, NS, VS, SC and CS (plus S in the codebook). PP is phrase-like; NS, VS, CS and SC are sentence- or clause-like; never treat them all as phrase types. Constituents nest and overlap; do not flatten them. The count 116 is constituent records, not reviewed phrases, and 66 clause-like candidates are not 66 validated clauses.
5. The rel_en column of RelLabels.csv contains relation codes such as cond, rslt and circ, not English glosses; use rel_ar for the Arabic description and preserve the source code.
6. The VerbMood sheet's English descriptions for JUS and SUBJ are reversed relative to the Arabic descriptions; correct them for display only and preserve raw values.

7.3 Feature-reading rules (QAC and EQTB adapters)

- Underscore, hyphen and blank placeholders count as missing, not unknown.
- MOOD:SUBJ and MOOD:JUS normalize to SUBJ and JUS for lookup only; SP:kaAn, SP:<in~ and SP:kaAd lose the SP: prefix for lookup only.
- Original raw values are always preserved; lookups never mutate the source files.

7.4 Surah 67 alignment facts (Package A production run)

In the Surah 67 run, 529 of 529 QAC morphology rows matched coordinate-bearing EQTB rows exactly; 54 EQTB implicit tokens were excluded from direct matching, and there were 0 duplicate location keys. The run produced 116 constituents (PP 50, NS 17, VS 44, SC 3, CS 2), 560 constituent-token memberships and 583 dependency records; 171 of 171 lemmas and 128 of 128 roots were found in the EQTB lexicons, which is a lookup check, not proof of linguistic correctness.

8. Annotation process

8.1 Generation pipeline

generate_spec.py batches roughly 350 words per call, writes per-surah annotation files (ayahs plus a status block with status, warnings, model, prompt_hash and depth), validates every reply (tiling, JSON shape, quotation rules, script checks), retries failures, marks survivors needs_human, logs token usage, and is resumable because finished ayahs are skipped. It has been tested offline only: no live API call has ever been made.

8.2 Anti-circularity (binding)

The annotating model sees only: numbered Arabic words, word glosses in English and Bangla, the two reference translations, CONVENTIONS.md and glossary.json. It never sees tags, sentence ids, constituent spans or mined rules. Tags and sentence ids attach afterwards through adapters into the canonical store's evidence and reference tables. Rules are mined only after the dataset exists. Translation memory flags conflicts only.

8.3 Review workflow

Generator output enters the canonical store as candidate, or pending if validation failed. Human review promotes items to reviewed; accepted releases mark accepted; unresolved items become uncertain or rejected. The full-Quran run reviews all flagged ayahs plus a random sample of about five percent. Bangla requires review by a Bangla reader; known draft issues are: the verb-first negative in 67:3; a clumsy rendering in 67:20; stiff Arabic order in 67:15 and 67:8; the dual word karretayni in 67:4:1 rendered with the informal Bangla for twice against Zakaria's second-time; and the resumption phrase wa-huwa rendered everywhere as the respectful and-he except in 67:4 where it is and-it-will-be, referring to the vision (English aliases it; the Bangla alias is still missing).

8.4 QA harness (merged)

1. Treebank cross-checks: sentence-start versus clause-start disagreements, and PP spans split by phrase boundaries. Al-Mulk baseline: 53 of 54 treebank sentence starts agree with clause starts, 0 of 50 PP spans are split, and there is one disagreement at 67:23:11, where the treebank starts a sentence at the words for how little you give thanks while the annotation kept little-indeed as one phrase; adjudicate this during the overlay proof of concept.
2. Same-Arabic-different-translation-key flags.
3. Mutashabihat comparison, sequence-based: repetition counting must use sequences of consecutive phrases (for example, the first four words of 67:25 repeat across six surahs but span three annotation phrases), not only whole phrases.
4. Tiling asserts and the QUL-versus-QAC root agreement assert, both already present in build_dataset.py.
5. Per-surah key-alignment validation across QPC, QUL and treebank (Package B's proposal, now a Phase 1 exit criterion).
6. Provenance completeness: every reviewed or accepted unit carries at least one evidence reference and a review record.
7. No-copied-source audit: the canonical store must contain Arabic strings only in the glossary table, and unit translations must not embed source rows.

9. What exists today

9.1 Package A assets

Package A contributes the six EQTB auxiliary files (verified intact), the Surah 67 dataset summary JSON quoted in Section 7.4, the eight-table schema proposal that starts the merged schema, the source registry and taxonomy templates, and two exploratory prototype scripts that are useful for parsing logic but are explicitly not the final architecture; their enriched CSV outputs must never be used as the annotation dataset.

9.2 Package B assets

Package B contributes the Al-Mulk annotation spec (30 ayahs; English finalised and normalised; Bangla is a draft pending review by a Bangla reader), CONVENTIONS.md with three drafted worked examples that must be replaced by reviewed ones, a ten-entry starter glossary, build_dataset.py (verified working: prints 333 words, 213 phrases, 86 clauses), analyze.py for pandas-based frequency and rule mining, convert_treebank.py (run successfully by the user; the output files still need moving into quranic-treebank/data/), the OpenRouter generator (dry-run tested only), Surah 67 source slices, and the tag legend files.

9.3 Mined hypotheses from Al-Mulk

These rules were measured on the 333-word Al-Mulk annotation, made before the tags were examined; the sample is small and the rules are to be re-measured on the whole Quran.

Table 4 Mined segmentation hypotheses (Al-Mulk pilot)

Rule	Result
Previous word carries a waqf mark, then clause start	15/15 (6 major, 9 minor marks)
Resumption prefix starts a clause	10/10
Negative or conditional particle starts a clause	12/12
Interrogative starts a clause	4/6
Coordinating conjunction starts a clause	9/17
Circumstantial marker starts a clause	1/3
Definite (al-) word never starts a clause	38/38
Bare preposition stays with the next word	22/22
Relative pronoun after a non-verb joins the previous word	9/10
Clause length 2-6 words	95%
Phrase length 1-3 words	98%
Phrases consisting of a single word	59%
Most repeated phrases: قُلْ x6; وَهُوَ x4; إِنْ x3; هُوَ ٱلَّذِي x3	(see left column)
Only repeated clause: قُلْ أَرَيْتُمْ x2	(see left column)

10. Merged canonical schema

The eight tables proposed by Package A are retained and extended. The merged schema adds translation fields and tiling spans to the unit table and introduces a glossary table.

Table 5 Canonical store tables

Table	Disposition in the merge
source_datasets	Kept. One row per source dataset with key, version, URL, licence and checksum. Add the QUL files and Mutashabihat as first entries.
annotation_units	Kept and extended: add translation_en, translation_bn, translation_key_en, translation_key_bn and tiling fields (ayah, word_start_key, word_end_key, display_order). unit_level stays phrase or clause.

Table	Disposition in the merge
unit_span_segments	Kept. Tiling base units get one segment each with QPC key endpoints; overlay spans may use multiple ordered segments.
unit_source_refs	Kept. Roles: exact_entity (reuse an EQTB constituent), evidence, candidate, derived_from, cross_check.
unit_relationships	Kept (typed and directional).
unit_classifications	Kept (six dimensions, taxonomy version, review status, confidence).
annotation_evidence	Kept (points to source records; never duplicates them).
annotation_reviews	Kept (accept, reject, revise, uncertain; reviewer and note).
glossary	New. From Package B's glossary.json: Arabic, English and Bangla term triples, versioned.

11. Roadmap

Phase 0 (done) and outstanding licence work

The architecture is frozen by this document. Outstanding from Phase 0: verify licence and version metadata for every source file.

Phase 1 (next, agreed): source registry, read-only adapters, resolvers

Build source_registry.yaml with dataset key, version or release, URL, licence identifier and file checksum for every source. Implement read-only adapters for the QUL word list, QUL ayah translations, QUL word glosses, QUL morphology databases, the EQTB main file (UTF-16 tab) and its six auxiliary files, QAC morphology, the converted treebank JSON, and the Mutashabihat files, applying the feature-reading rules of Section 7.3. Implement resolvers: QPC key to EQTB (sentence_id, token_id) or tid; QPC key to QAC s:a:w:segment; the implicit-token policy (never invent coordinates); and the constituent span resolver using sentence_id plus inclusive constituents_loc ranges. Exit criteria: given any stored reference, the tool retrieves the original row on demand; checksums before and after prove sources unmodified; per-surah key-alignment validation passes across QPC, QUL and treebank.

Phase 2: canonical store and round-trip

Implement the merged schema of Section 10; convert the Al-Mulk spec into the store as the first test data; re-render the Surah 67 export from store plus sources and diff it against the existing mulk.json (it must match modulo declared differences); run the no-copied-source audit; decide on flat JSONL exports for analysis.

Phase 3: generator tuning on Al-Mulk

Run the generator in dry-run mode, then a live Surah 67 run with a chosen model; write a comparison script against the Al-Mulk gold annotation measuring boundary agreement, phrase counts and translation_key consistency; tune CONVENTIONS.md; replace the three drafted worked examples with reviewed ones; grow the glossary; have the Bangla reviewed by a Bangla reader.

Phase 4: pilot surahs and the linguistic overlay proof of concept

Pilot about five varied surahs interactively: 1, 12, 112, 2 (especially 2:255 and 2:282) and 67. In parallel run the overlay proof of concept on Surah 67: a reuse-versus-create inventory that references exact EQTB constituents and creates units only for missing spans; first classifications with evidence and reviews; adjudicate the 67:23:11 disagreement.

Phase 5: QA harness

Implement the seven checks of Section 8.4 as one script suite.

Phase 6: full run and review

Run all 114 surahs; review all flagged ayahs plus a random sample of about five percent; finalise annotation guidelines and adjudication rules; if a second annotator is available, measure inter-annotator consistency on a sample.

Phase 7: mining and release

Mine rules and frequencies on reviewed and accepted ayahs only; version the dataset; complete the licence and attribution review before any redistribution.

Phase 8: application integration

Build phrase and clause browsing and search; resolve source data on demand; filter by type, function, root, lemma, surah and verse; display source citations and provenance; mark candidate and uncertain states clearly.

12. Open items and not verified

- Live OpenRouter calls have never been made; the request format assumes the OpenAI-style chat-completions API.
- The jq examples in Package B's README are untested.
- Bangla quality is unreviewed; known issues are listed in Section 8.3.

- Token and cost estimates are rough: the whole Quran is about 1.3 to 1.8 million read tokens plus 0.6 to 0.9 million written tokens, plus roughly 2,000 to 3,000 fixed prompt tokens per batch.
- Licences are pending verification: EQTB data licence, QAC data-use terms, QUL terms.
- The full Quranic.csv and the QAC morphology file are not in either package; supply from original downloads.
- The converted treebank files need moving into quranic-treebank/data/.
- The waqf_strength field idea (major versus minor pause) remains open.
- The taxonomy template is a scaffold; the phrase and clause taxonomies need grounding in a chosen Arabic grammatical framework (Phase 4).
- Dependency relations: storage is deferred; EQTB relation labels still resolve at render time for display.

Appendix A. Status mapping

Table 6 Status mapping from Package B to the canonical vocabulary

Canonical	Package B origin	Meaning
candidate	auto	Generated automatically; never shown as gold
pending	needs_human	Awaiting review (validation failed, flagged, or queued)
uncertain	—	Plausible but unresolved
reviewed	reviewed	Checked by a human
accepted	—	Approved into the released set
rejected	—	Reviewed and not accepted

Appendix B. Rendered export field provenance

The rendered per-surah JSON stays in Package B's shape. This table states, per field group, where the value comes from at render time and whether it is stored in the canonical store.

Table 7 Field provenance in the rendered export (cache)

Field group	Level	Resolved from at render	In canonical store?
text, text_norm, waqf_mark	word	QUL word list	No
root, lemma, stem, gloss_en/bn	word	QUL morphology databases / wbw glosses	No

Field group	Level	Resolved from at render	In canonical store?
pos, cat, prefix_tags, features, definite, suffix_pron, qac_lemma	word	QAC / treebank JSON	No
boundary_before, is_clause_start, is_phrase_start, prev_word_waqf	word	Derived from the store	No (derivable)
word/phrase/clause spans, translations, translation_keys	unit	—	Yes (the new annotation)
reference_translations, translations_phrased	ayah	QUL translations + units	No (resolved)
mutashabihat, matching_ayah	ayah	Mutashabihat files	No (resolved)
pattern, first_cat/last_cat, tag lookups	derived	Codebooks (auxiliary files)	No (resolved)

Appendix C. Working commands

Commands from Package B; the build is verified working, the generator is dry-run only.

```
cd mulk-dataset && python3 scripts/build_dataset.py
    # verified: prints 333 words 213 phrases 86 clauses
python3 scripts/analyze.py > data/analysis_report.txt   # needs pandas
python3 scripts/convert_treebank.py --qpc quran-script/wbw/qpc-hafs-word-by-
word.json
    # run once; move outputs into quranic-treebank/data/
export OPENROUTER_API_KEY=...
python3 scripts/generate_spec.py --data-root /path/to/qul --surah 67 --dry-run
python3 scripts/generate_spec.py --data-root /path/to/qul --surah 67 --model
<slug>
python3 scripts/generate_spec.py --data-root /path/to/qul --surahs 1-114 --
model <slug>
python3 scripts/generate_spec.py ... --retry-failed
# Package A legacy prototype (exploratory only):
python prototype/build_surah_mulk_dataset.py --eqtb data/Quranic.csv \
  --qac data/quranic-corpus-morphology-0.4.txt --aux-dir reference/eqtb_aux \
  --out outputs/surah_67_full
```

Appendix D. Where the original assets live

Table 8 Asset map

Asset	Location
Package A (architecture docs, schema, EQTB auxiliary files, Surah 67 summary, prototypes)	handoff_extracted/ quranic_phrase_clause_handoff/

Asset	Location
Package B (mulk-dataset: annotations, scripts, data, sources, reference legends)	handoff2_extracted/mulk-dataset/
Original archives	upload/quranic_phrase_clause_handoff.tar.gz and upload/mulk-handoff.zip
This merged handoff	download/MERGED_HANDOFF.md plus this Word reading copy